

- 5 Some disorders are caused by mutations in the mitochondrial DNA and can be inherited.

Three-parent embryos have been developed to prevent the inheritance of these disorders.

Diagram 1 shows how three-parent embryos are made.

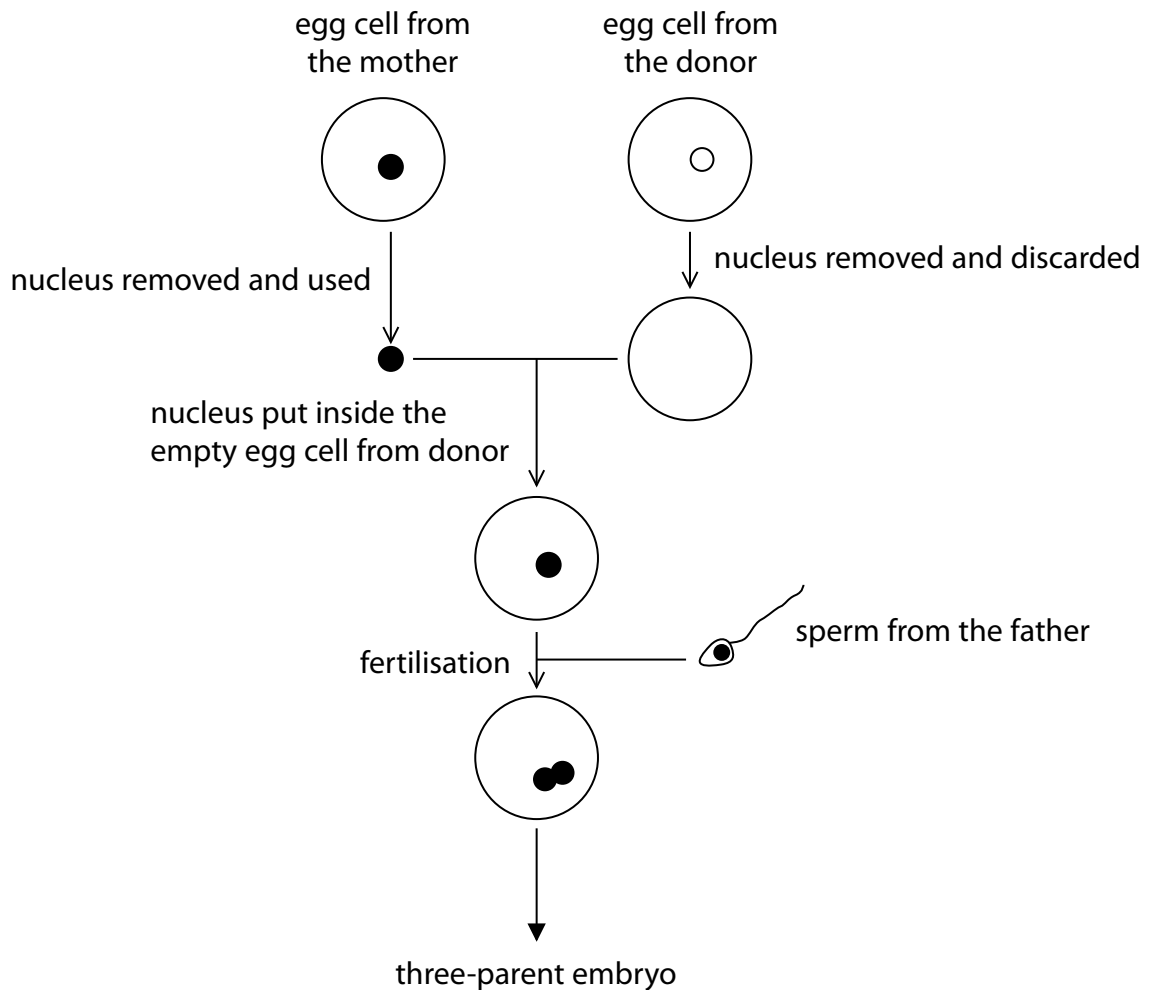


Diagram 1

- (a) The cells in the three-parent embryo are totipotent stem cells.

State what is meant by the term **totipotent stem cells**.

(2)

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(b) Diagram 2 shows the results of separating the DNA from the three-parent embryo, the mother and the father, using gel electrophoresis.

The bands from the DNA of the embryo are labelled from 1 to 12.

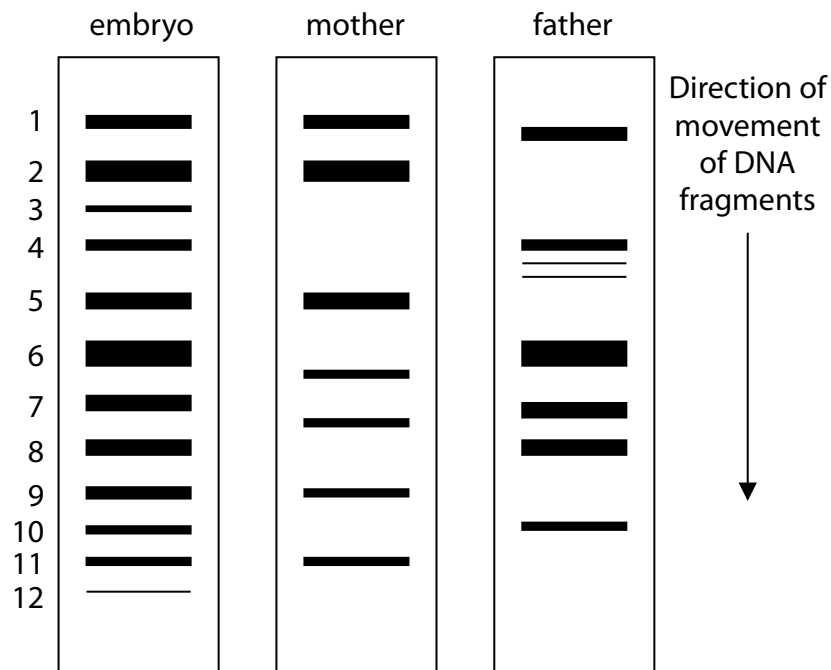


Diagram 2

(i) Which of the following is the reason for the movement of the DNA fragments?

(1)

- ☐ **A** negatively-charged fragments move towards the negative end of the gel
- ☐ **B** negatively-charged fragments move towards the positive end of the gel
- ☐ **C** positively-charged fragments move towards the negative end of the gel
- ☐ **D** positively-charged fragments move towards the positive end of the gel

(ii) Which band on the gel contains the heaviest DNA fragments?

(1)

☐ **A** 1

☐ **B** 3

☐ **C** 6

☐ **D** 9

*(iii) Explain the banding pattern of this three-parent embryo. Use information in Diagram 1 and Diagram 2 to support your answer.

(6)

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(Total for Question 5 = 10 marks)